Definition

For real-valued nets f and g,

$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|$ iff $f(x) \in O(g(x))$ and $|g(x)|$ is an asymptotic upper bound of $|f(x)|$.

$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M|g(x)|$ iff $f(x) \in \Omega(g(x))$ and $|g(x)|$ is an asymptotic lower bound of $|f(x)|$.

$\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|$ iff $f(x) \in \Theta(g(x))$.

$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M|g(x)|$ iff $f(x) \in o(g(x))$.

$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M|g(x)|$ iff $f(x) \in \omega(g(x))$.

$\in$ is also denoted as $=$ for them.

Limit

If $\exists k \in \mathbb{R} : x \geq k \implies g(x) \neq 0$,

$\limsup \left|\frac{f(x)}{g(x)}\right| < \infty$ iff $f(x) \in O(g(x))$.

Proof. $|f(x)| \leq M|g(x)|, \quad x \geq T; \left|\frac{f(x)}{g(x)}\right| \leq M, \quad x \geq T;$ $\sup \left|\frac{f(x)}{g(x)}\right| \leq M, \quad x \geq T; \limsup \left|\frac{f(x)}{g(x)}\right| \leq M < \infty;$ □

$\liminf \left|\frac{f(x)}{g(x)}\right| > 0$ iff $f(x) \in \Omega(g(x))$.

Proof. $|f(x)| \geq M|g(x)|, \quad x \geq T; \left|\frac{f(x)}{g(x)}\right| \geq M, \quad x \geq T;$ $\inf \left|\frac{f(x)}{g(x)}\right| \geq M, \quad x \geq T; \liminf \left|\frac{f(x)}{g(x)}\right| \geq M > 0;$ □

$\infty > \limsup \left|\frac{f(x)}{g(x)}\right| \geq \liminf \left|\frac{f(x)}{g(x)}\right| > 0$ iff $f(x) \in \Theta(g(x))$.

Proof. $|f(x)| \geq M_1|g(x)|, \quad x \geq T; \left|\frac{f(x)}{g(x)}\right| \geq M_1, \quad x \geq T;$ $\inf \left|\frac{f(x)}{g(x)}\right| \geq M_1, \quad x \geq T; \liminf \left|\frac{f(x)}{g(x)}\right| \geq M_1 > 0;$ $|f(x)| \leq M_2|g(x)|, \quad x \geq T; \left|\frac{f(x)}{g(x)}\right| \leq M_2, \quad x \geq T;$ $\sup \left|\frac{f(x)}{g(x)}\right| \leq M_2, \quad x \geq T; \limsup \left|\frac{f(x)}{g(x)}\right| \leq M_2 < \infty;$ □

$\lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = 0$ iff $f(x) \in o(g(x))$. $|f(x)| < M|g(x)|;$ $\left|\frac{f(x)}{g(x)}\right| < M; \lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = 0;$

$\lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = \infty$ iff $f(x) \in \omega(g(x))$. $|f(x)| > M|g(x)|;$ $\left|\frac{f(x)}{g(x)}\right| > M; \lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right| = \infty;$

Relation

For any fixed $g(x)$, $o(g(x))$, $\Theta(g(x))$, and $\omega(g(x))$ are mutually exclusive.

Proof. Prove by contradiction. If $f \in o(g(x)) \cap \Theta(g(x))$, $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M|g(x)|;$ $\exists M_1, M_2 > 0, U \in \mathbb{R} : x \geq U \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|$; then set $M = M_1$, for $x \geq \max\{T, U\}$ $M_1|g(x)| \leq |f(x)| < M_1|g(x)|$.; which can't happen. If $f \in \omega(g(x)) \cap \Theta(g(x))$, $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M|g(x)|;$ $\exists M_1, M_2 > 0, U \in \mathbb{R} : x \geq U \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|$; then set $M = M_2$, for $x \geq \max\{T, U\}$ $M_2|g(x)| \geq |f(x)| > M_2|g(x)|$.; which can't happen. If $f \in o(g(x)) \cap \omega(g(x))$, $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M|g(x)|;$ $\forall M > 0 : \exists U \in \mathbb{R} : x \geq U \implies |f(x)| > M|g(x)|$; then set any fixed M, $M|g(x)| < |f(x)| < M|g(x)|$.; which can't happen. □

For any fixed $g(x)$, $o(g(x)) \subseteq O(g(x))$;

Proof. Peak any $M > 0$. □

For any fixed $g(x)$, $\omega(g(x)) \subseteq \Omega(g(x))$;

Proof. Peak any $M > 0$. □

For any fixed $g(x)$, $\Theta(g(x)) = O(g(x)) \cap \Omega(g(x))$;

Proof. Prove $\Theta(g(x)) \subseteq O(g(x)) \cap \Omega(g(x))$; $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|;$ $\exists N > 0, U \in \mathbb{R} : x \geq U \implies |f(x)| \geq N|g(x)|$; then $\exists M, N > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies N|g(x)| \leq |f(x)| \leq M|g(x)|$;

Prove $\Theta(g(x)) \supseteq O(g(x)) \cap \Omega(g(x))$; $\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|$; then $\exists M_2 > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M_2|g(x)|$; $\exists M_1 > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M_1|g(x)|$; □

For any fixed $g(x)$, $O(g(x)) \setminus (o(g(x)) \cup \Theta(g(x))) \neq \varnothing$;

Proof. Counterexample: $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}; g(x) = 1;$ $f \in O(g(x))$: $\exists 1 > 0, 1 \in \mathbb{R} : x \geq 1 \implies |f(x)| \leq |g(x)|;$ $f \notin o(g(x))$: $\forall 1 > 0, T \in \mathbb{R} : \exists x \geq T, x \in \mathbb{Q} : |f(x)| = |g(x)|;$ $f \notin \Theta(g(x))$: $\forall M_1 > 0, x \in \mathbb{R}, x \notin \mathbb{Q} : M_1|g(x)| = M_1 > |f(x)| = 0;$ □

For any fixed $g(x)$, $\Omega(g(x)) \setminus (\omega(g(x)) \cup \Theta(g(x))) \neq \varnothing$;

Proof. Counterexample: $f(x) = 1; g(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases};$ $f \in \Omega(g(x))$: $\exists 1 > 0, 1 \in \mathbb{R} : x \geq 1 \implies |f(x)| \geq |g(x)|;$ $f \notin \omega(g(x))$: $\forall 1 > 0, T \in \mathbb{R} : \exists x \geq T, x \in \mathbb{Q} : |f(x)| = |g(x)|;$ $f \notin \Theta(g(x))$: $\forall M_2 > 0, x \in \mathbb{R}, x \notin \mathbb{Q} : M_2|g(x)| = 0 < |f(x)| = 1;$ □

$f(x) \in O(g(x)) \iff g(x) \in \Omega(f(x))$;

Proof. $\implies$: $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|$; then $\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \geq \frac{1}{M}|g(x)|$; $\impliedby$: $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M|g(x)|$; then $\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \leq \frac{1}{M}|g(x)|$; □

$f(x) \in o(g(x)) \iff g(x) \in \omega(f(x))$;

Proof. $\implies$: $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M|g(x)|$; then $\forall \frac{1}{M} > 0 : \exists T \in \mathbb{R} : x \geq T \implies |g(x)| > \frac{1}{M}|f(x)|$; $\impliedby$: $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M|g(x)|$; then $\forall \frac{1}{M} > 0 : \exists T \in \mathbb{R} : x \geq T \implies |g(x)| < \frac{1}{M}|f(x)|$; □

For any fixed $g(x)$, $\exists f(x) : f(x) \notin O(g(x)) \cup \Omega(g(x))$;

Proof. $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ x, & x \notin \mathbb{Q} \end{cases}, g(x) = \begin{cases} x, & x \in \mathbb{Q} \\ 1, & x \notin \mathbb{Q} \end{cases};$ For $x \in \mathbb{R}\setminus\mathbb{Q}$, $\nexists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|$; For $x \in \mathbb{Q}$, $\nexists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M|g(x)|$; □

Binary relation

$\Theta(g(x))$ is an equivalence class.

Proof. Transitive: $\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|;$ $\exists N_1, N_2 > 0, U \in \mathbb{R} : x \geq U \implies N_1|h(x)| \leq |g(x)| \leq N_2|h(x)|$; then $\exists M_1N_1, M_2N_2 > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies M_1N_1|h(x)| \leq |f(x)| \leq M_2N_2|h(x)|$; Reflexive: $f(x) \in \Theta(f(x))$; $\exists 1, 1 > 0, a \in \mathbb{R} : x \in \mathbb{R} \implies 1|f(x)| \leq |f(x)| \leq 1|f(x)|$; Symmetric: $\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1|g(x)| \leq |f(x)| \leq M_2|g(x)|$; then $\exists \frac{1}{M_2} \cdot \frac{1}{M_1} > 0, T \in \mathbb{R} : x \geq T \implies \frac{1}{M_2}|f(x)| \leq |f(x)| \leq \frac{1}{M_1}|f(x)|$; □

$O(g(x))$ is partial-ordered set with Θ as $=$.

Proof. Transitive: $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|;$ $\exists N > 0, U \in \mathbb{R} : x \geq U \implies |g(x)| \leq N|h(x)|$; then $\exists MN > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies |f(x)| \leq MN|h(x)|$; Reflexive: $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|$; then $\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \leq \frac{1}{M}|f(x)|$; Antisymmetric: $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M|g(x)|;$ $\exists N > 0, U \in \mathbb{R} : x \geq U \implies |g(x)| \leq N|f(x)|$; then $\exists M, \frac{1}{N} > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies \frac{1}{N}|g(x)| \leq |f(x)| \leq M|g(x)|$; □

$\Omega(g(x))$ is partial-ordered set with Θ as $=$.

Proof. Similar to that of O. □

$o(g(x))$ is a strictly partially-ordered set.

Proof. Asymmetry: $f(x) \in o(g(x)) \iff g(x) \in \omega(f(x)) \implies g \notin o(f(x))$; Transitivity: $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M|g(x)|;$ $\exists U \in \mathbb{R} : x \geq U \implies |g(x)| < |h(x)|$; then $\forall M > 0 : \exists \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies |f(x)| < M|h(x)|$; □

$\omega(g(x))$ is a strictly partially-ordered set.

Proof. Similar to that of o. □

Integral test (for convergence) or Maclaurin – Cauchy test

Suppose $f(x) : [a, \infty) \to \mathbb{R}$ for some $a \in \mathbb{N}$ is monotone and $\lim_{x\to\infty} f(x) = 0$. Then $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent iff $\sum_{k=a}^{\infty} f(k)$ is convergent. And,

If $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent and $f(x)$ is monotone decreasing, for any $n \in \mathbb{N}_{\geq a}$, $\int_n^\infty f(x)\,\mathrm{d}x \leq \sum_{k=n}^{\infty} f(k) \leq f(n) + \int_n^\infty f(x)\,\mathrm{d}x$.; called remainder estimate for integral test. Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x)\,\mathrm{d}x \leq \int_{n-1}^\infty f(x)\,\mathrm{d}x$.

If $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent and $f(x)$ is monotone increasing, for any $n \in \mathbb{N}_{\geq a}$, $\int_n^\infty f(x)\,\mathrm{d}x \geq \sum_{k=n}^{\infty} f(k) \geq f(n) + \int_n^\infty f(x)\,\mathrm{d}x$.; called remainder estimate for integral test. Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x)\,\mathrm{d}x \geq \int_{n-1}^\infty f(x)\,\mathrm{d}x$.

Proof. Consider the case where $f(x)$ is monotone increasing. The case where $f(x)$ is monotone decreasing can be proven similarly. First consider the case where $\int_a^\infty f(x)\,\mathrm{d}x$ is convergent. This implies that $f(x)$ is non-positive. Thus, for any fixed $n \in \mathbb{N}_{\geq a}$, $\int_n^{n+1} f(x)\,\mathrm{d}x \geq f(n)$.; and for any $k \geq n+1$, $\int_k^{k+1} f(x)\,\mathrm{d}x \geq f(k) \geq \int_{k-1}^{k} f(x)\,\mathrm{d}x$.; For any $N \in \mathbb{N}_{\geq n}$, summing over all $k \in \mathbb{N}_{\geq n}$, we obtain, $\int_n^{N+1} f(x)\,\mathrm{d}x \geq \sum_{k=n}^{N} f(k) \geq f(n) + \int_n^N f(x)\,\mathrm{d}x$.; Take limit $\lim_{N\to\infty}$, we obtain $\int_n^\infty f(x)\,\mathrm{d}x \geq \sum_{k=n}^{\infty} f(n) \geq f(n) + \int_n^\infty f(x)\,\mathrm{d}x$.; Second, consider the case where $\int_a^\infty f(x)\,\mathrm{d}x$ is divergent. Since $f(x)$ is monotone. Either $\int_a^\infty f(x)\,\mathrm{d}x = -\infty$ or $\int_a^\infty f(x)\,\mathrm{d}x = \infty$. For the former, the supremum of $f(x)$ is non-positive, and $\sum_{k=a}^{\infty} f(k) = -\infty$. For the latter, there exists $x_0 > a$ such that $f(x) > 0$ for all $x \geq x_0$, and thus $\sum_{k=a}^{\infty} f(k) = \infty$. □

p-test (for improper integral)

For any fixed $a > 0$, $\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x$ converges iff $p > 1$, and if so, $\int_a^\infty \frac{1}{x^p}\,\mathrm{d}x = \frac{1}{(p-1)\,a^{p-1}}$.; $\int_0^a \frac{1}{x^p}\,\mathrm{d}x$ converges iff $p < 1$, and if so, $\int_0^a \frac{1}{x^p}\,\mathrm{d}x = -\frac{1}{(p-1)\,a^{p-1}}$.;

Proof. $\int \frac{1}{x^p}\,\mathrm{d}x = \begin{cases} -\frac{1}{(p-1)\,x^{p-1}}, & p \neq 1 \\ \ln(x), & p = 1 \end{cases};$ $\forall p < 1 : \nexists \lim_{x\to\infty} -\frac{1}{(p-1)\,x^{p-1}}; \quad \nexists \lim_{x\to\infty} \ln(x); \quad \forall p > 1 : \lim_{x\to\infty} -\frac{1}{(p-1)\,x^{p-1}} = 0 \quad \int_a^\infty \frac{1}{x^p}\,\mathrm{d}x = \frac{1}{(p-1)\,a^{p-1}};$ $\forall p > 1 : \nexists \lim_{x\to 0^+} -\frac{1}{(p-1)\,x^{p-1}}; \quad \nexists \lim_{x\to 0^+} \ln(x); \quad \forall p < 1 : \lim_{x\to 0^+} -\frac{1}{(p-1)\,x^{p-1}} = 0 \quad \int_0^a \frac{1}{x^p}\,\mathrm{d}x = -\frac{1}{(p-1)\,a^{p-1}};$ □

Generalized p-test (for improper integral)

Define iterated logarithmic function $L_n(x)$ as $L_0(x) = x, \quad x \in \mathbb{R}_{>0}$; $L_n(x) = \ln\left(L_{n-1}(x)\right), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}$.; For any fixed $a > 1, k \in \mathbb{N}$, $\int_a^\infty \frac{1}{\left(\prod_{i=0}^{k-1} L_i(x)\right)\left(L_k(x)\right)^m}\,\mathrm{d}x$; converges iff $m > 1$.

Proof. Let $u = L_k(x)$; $\mathrm{d}u = \frac{\mathrm{d}x}{\prod_{i=0}^{k-1} L_i(x)}$; $\int \frac{1}{\left(\prod_{i=0}^{k-1} L_i(x)\right)\left(L_k(x)\right)^m}\,\mathrm{d}x = \int \frac{1}{u^m}\,\mathrm{d}u$; By p-test, the statement holds. □

For any fixed $0 < a < 1$, $\int_0^a \frac{1}{x(\ln x)^m}\,\mathrm{d}x$; converges iff $m < 1$.

Proof. By substitution of $u = \ln x$. □

Generalized p-test for series or generalized p-series test

Define iterated logarithmic function $L_n(x)$ as $L_0(x) = x, \quad x \in \mathbb{R}_{>0}$; $L_n(x) = \ln\left(L_{n-1}(x)\right), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}$.; For any fixed $a \in \mathbb{N}$, $\sum_{n=a}^{\infty} \frac{1}{\left(\prod_{i=0}^{k-1} L_i(n)\right)\left(L_k(n)\right)^m}$; converges iff $m > 1$.

Proof. By integral test and generalized p-test for improper integral, the statement holds. □

Generalized Master Theorem

For a positive valued net $T(n)$ defined on $n = b^h$ for any $h \in \mathbb{N}_0$ as $T(n) = aT\left(\frac{n}{b}\right) + f(n)$; for some constants $a > 0$ and $b > 1$, and non-negative-valued function $f(n)$,

If for some $c < \log_b a$, $f(n) = O\left(n^c\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\right)$.;

Define iterated logarithmic function $L_n(x)$ as $L_0(x) = x, \quad x \in \mathbb{R}_{>0}$; $L_n(x) = \ln\left(L_{n-1}(x)\right), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}$.;

If for some positive integers k and positive number m, $f(n) = \Theta\left(n^{\log_b a}\left(\prod_{i=1}^{k-1}\left(L_i(n)\right)^{-1}\right)\left(L_k(n)\right)^{m-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\left(L_k(n)\right)^m\right)$.;

If for some positive integer k, $f(n) = \Theta\left(n^{\log_b a}\prod_{i=1}^{k}\left(L_i(n)\right)^{-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}L_{k-1}(n)\right)$.;

If for some positive integers k and some positive number m, $f(n) = \Theta\left(n^{\log_b a}\left(\prod_{i=1}^{k-1}\left(L_i(n)\right)^{-1}\right)\left(L_k(n)\right)^{-m-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\right)$.;

If $f(n) = \Omega\left(n^c\right)$ for some $c > \log_b a$, and the regularity condition that $\exists m \in \mathbb{R}, k < 1$ such that $n \geq m$ implies $af\left(\frac{n}{b}\right) \leq kf(n)$; is satisfied, then $T(n) = \Theta(f(n))$.;

Proof. Let $s = \log_b a$, $n = b^h$, and $f(x) = x^s g(x)$. At level j, there are a^j nodes, each of size $\frac{n}{b^j}$. Thus, the total non-recursive work at level j is $a^j f\left(\frac{n}{b^j}\right)$.; Hence $T(n) = \sum_{j=0}^{h-1} a^j f\left(\frac{n}{b^j}\right) + n^s T(1) = \sum_{j=0}^{h-1} b^{sj} f\left(\frac{b^h}{b^j}\right) + \Theta\left(n^s\right) = n^s \sum_{j=0}^{h-1} g\left(b^{h-j}\right) + \Theta\left(n^s\right) = n^s \sum_{r=1}^{h} g\left(b^r\right) + \Theta\left(n^s\right)$; Case 1: If for some $c < \log_b a$, $f(n) = O\left(n^c\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\right)$.; Case 2: $L_k\left(b^x\right) = \Theta\left(L_{k-1}(x)\right)$; $f(n) = \Theta\left(n^{\log_b a}\left(\prod_{i=1}^{k-1}\left(L_i(n)\right)^{-1}\right)\left(L_k(n)\right)^{m-1}\right)$; $g\left(b^r\right) = \Theta\left(\left(\prod_{i=1}^{k-2}\left(L_i(r)\right)^{-1}\right)\left(L_{k-1}(r)\right)^{m-1}\right)$; By generalized p-test for series: Sub-case 1 $m > 0$: $T(n) = \Theta\left(n^{\log_b a}\left(L_k(n)\right)^m\right)$.; Sub-case 2 $m = 0$: $T(n) = \Theta\left(n^{\log_b a}L_{k-1}(n)\right)$.; Sub-case 3 $m = 0$: $T(n) = \Theta\left(n^{\log_b a}\right)$.; Case 3: If $f(n) = \Omega\left(n^c\right)$ for some $c > \log_b a$, and the regularity condition that $\exists m \in \mathbb{R}, 0 < k < 1$ such that $n \geq m$ implies $af\left(\frac{n}{b}\right) \leq kf(n)$; is satisfied, then $af\left(\frac{n}{b}\right) \leq kf(n)$; $a^j f\left(\frac{n}{b^j}\right) \leq a^{j-1}kf\left(\frac{n}{b^{j-1}}\right) \leq k^j f(n)$; $T(n) = \sum_{j=0}^{h-1} a^j f\left(\frac{n}{b^j}\right) + n^s T(1) = f(n) + \sum_{j=1}^{h-1} O\left(k^j f(n)\right) + \Theta\left(n^s\right)$; $\sum_{j=1}^{\infty} k^j f(n) = \frac{f(n)}{1-k}$; $T(n) = f(n) + O(f(n)) + \Theta\left(n^s\right) = \Theta(f(n))$; □

Terms:

h is called recursion level.

$\log_b a$ is called critical exponent c_{crit}.

$T(1)$ is called leaf cost.

At each node, the first term is called recursive work, and the second term is called non-recursive work.

For original master theorem, the second case is

If for some positive integers m, $f(n) = \Theta\left(n^{\log_b a}(\ln n)^{m-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}(\ln n)^m\right)$.;

If $f(n) = o\left(n^{\log_b a}(\ln n)^{-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\ln(\ln n)\right)$.;

If for some positive integers m, $f(n) = \Theta\left(n^{\log_b a}(\ln n)^{-m-1}\right)$.; then $T(n) = \Theta\left(n^{\log_b a}\right)$.;

Stirling's approximation

$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n\left(1 + O\left(\frac{1}{n}\right)\right)$;

特徵方程式

定義：對一個給定 k 階常係數線性遞迴數列：

$$< a_n >: a_n = \sum_{i=1}^{k} c_i \cdot a_{n-i} + f(n)$$

，其特徵方程式為 $x^k - \sum_{i=1}^{k} c_i \cdot x^{k-i} = 0$，特證方程式的根稱其特徵根。

若一個給定 k 階常係數線性齊次遞迴數列 $< a_n >$ 有 a 相異特徵根 $q_1, q_2, q_3, \ldots, q_a$，其中第 i 個根 $(1 \leq i \leq a)$ 為 m_i 重根，則 $< a_n >$ 之一般式為：$a_n = \sum_{i=1}^{a}\left\{\sum_{j=1}^{m_i}\left(A_{i,j} \cdot n^{j-1}\right) \cdot q_i^{\,n}\right\}$;

若給定一個 k 階常係數線性非齊次遞迴數列：$< a_n >: a_n = \sum_{i=1}^{k} c_i \cdot a_{n-i} + f(n)$; 則令：$a_n = a'_n + a''_n$; 使得 $< a'_n >$ 為 k 階常係數線性齊次遞迴數列，$< a''_n >$ 為 k 階常係數線性非齊次遞迴數列且一次項同 $< a'_n >$ 且 $f(n)$ 項同 $< a_n >$。$a''_n = g(n)$, $\deg(g(n)) \leq \deg(f(n))$; 解出 k 階常係數線性非齊次遞迴數列 $< a'_n >$ 與 $g(n)$，即可解出 $< a_n >$。

Python and DS

Python operator precedence order: ( ) > ** > +x -x ~x (bitwise NOT) > * / // % > + - > « » (preserve sign) > & (bitwise AND) >^(bitwise XOR) > | (bitwise OR) > == != > >= < <= (value comparison) `is` `is not` (identity) `in` `not in` > `not` > `and` > `or`

Stack is last-in-first-out (LIFO) O(1) with push as inputting an element, pop as removing and returning the last inputted element, top as peaking the last inputted element, top as the last inputted element, and bottom as the first inputted element. list is a stack with append as push, pop as pop, and [-1] as top. SinglyLinkedList is a stack with prepend as push, pop_front as pop, and peak_front as top.

Queue is first-in-first-out (FIFO) O(1) with enqueue as inputting an element, dequeue as removing and returning the first inputted element, peak as peaking the first inputted element, front as the first inputted element, and back or rear as the last inputted element. SinglyLinkedList is a queue with append as enqueue, pop_front as dequeue, and peak_front as top.

To keep track of the extreme element of a stack or queue in O(1) time with O(n) extra space. One can maintain the original stack of elements and an extra stack of extrema. When an element is pushed to the original stack, it is compared with the top of the extrema stack and the extreme one is pushed to the extrema stack (for the first element, it is pushed to the extrema stack). When an element is popped from the original stack, an element is popped from the extrema stack. This way, the top of the extrema stack is always the extreme of the elements in the original stack.

Doubly-ended queue (Deque) is both-end-in-out O(1) and is a stack and a queue.

An unordered list is a collection of items where each item holds a relative position with respect to the others, such as SinglyLinkedList and DoublyLinkedList.

An ordered list is a collection of items which are ordered based upon some underlying characteristic of the item, which can be implemented with a stack or a queue if O(1) pop of one end is needed and a deque if O(1) pop of both ends is needed.

```
bool(0)  # False, only 0 False, other
         numbers True
2**72  # int is of infinite precision,
       ** is power
pow(2, 72)  # same as 2**72
float("inf")  # infinite float
(1).__add__(2)  # 3, must use (1).
      since 1. is treated as decimal
3 // 2  # 1, floor quotient
3 // 2  # 1.5, quotient
divmod(3, 2)  # (1, 1), the pair (x //
              y, x % y)
abs(-2)  # 2
c = complex(1, 2)  # (1+2j), complex(re
     , im)
c.conjugate()  # (1-2j)
a = 0.5  # a == 0.5
type(True)  # <class 'bool'>
type(1)  # <class 'int'>
type(1.0)  # <class 'float'>
type(complex(1, 1))  # <class 'complex
     '>
a = None
a is None  # True
a = [1]
b = a
a == b  # True
a != b  # False
a is b  # True, i.e., id(a) == id(b)
a is not b  # False
b = a.copy()
a == b  # True
a is b  # False
c, d = a, b
c, d = d, c  # swap
3 if 2 > 1 else 2  # 3
sradius = input("Please enter the
          radius of the circle ")
# ...
# Please enter the radius of the circle
          (input here, say 1)
# sradius == 1
# print(obj[, obj ,...], sep=' ', end
       ='\n', file=sys.stdout, flush=
       False)
# obj: Arbitrary number of objects that
       can be converted to str. Will
       be converted to str and printed
       .
# sep: Separator between objects
# end: What to print at end
# file: An object with a write method
       to print to
# flush: True for flushed, False for
       buffered
print("a")  # a
print(1, [1])  # 1 [1]
print(1, 2, sep=",", end="1")  # 1,21<
      no_newline>
print(1, 2, sep=", ", end="")  # 1, 2<
      no_newline>




a = "a"
type(a)  # <class 'str'>
a += "b"  # a == 'ab'
"He" .__add__("llo")  # 'Hello'
str(1)  # '1'
"g" in "egg"  # True
"f" in "egg"  # False
len("Hello")  # 5
"Hello World".capitalize()  # 'Hello
          world', first capitalize, other
          lowercase
# str.center(width, fillchar=' '),
          Return centered in a string of
          length width. Padding is done
          using the specified fillchar (
          default is an ASCII space). The
           original string is returned if
           width is less than or equal to
           len(s).
"Python".center(10) #'  Python  '
"Python".center(10, "-")  #'--Python
          --'
"Python".center(4)  # 'Python'
# str.count(sub[, start[, end]])
"spam, spam, spam".count("spam")  # 3
"spam, spam, spam".count("spam", 5)  #
          2
"spam, spam, spam".count("spam", 5, 10)
          # 1
"spam, spam, spam".count("eggs")  # 0
"spam, spam, spam".count("") # 17
# str.endswith(suffix[, start[, end]]),
          Return True if the string ends
          with the specified suffix,
          otherwise return False. suffix
          can also be a tuple of suffixes
          to look for. With optional
          start, test beginning at that
          position. With optional end,
          stop comparing at that position
          . Using start and end is
          equivalent to str[start:end].
          endswith(suffix).
# suffix can also be a tuple of
          suffixes to look for.
"Python".endswith("on")  # True
"a tuple of suffixes".endswith(("at", "
          in"))  # False
"a tuple of suffixes".endswith(("at", "
          es"))  # True
"Python is amazing".endswith("is", 0,
          9)  # True
# str.startswith(prefix[, start[, end
          ]]), endswith but startswith
# str.expandtabs(tabsize=8)
"01\t012\t0123\t01234".expandtabs()  #
          '01  012 0123    01234'
"01\t012\t0123\t01234".expandtabs(4)  #
          '01  012 0123    01234'
```

```
# str.find(sub[, start[, end]]), Return
          the lowest index in the string
          where substring sub is found
          within the slice s[start:end].
          Optional arguments start and
          end are interpreted as in slice
           notation. Return -1 if sub is
          not found.
"spam, spam, spam".find("sp")  # 0
"spam, spam, spam".find("sp", 5)  # 6
# str.rfind(sub[, start[, end]]), find
          but highest index
# str.index(sub[, start[, end]]), Like
          find(), but raise ValueError
          when the substring is not found
          .
"spam, spam, spam".index("spam")  # 0
# 'spam, spam, spam'.index('eggs')  #
          ValueError: substring not found
# str.rindex(sub[, start[, end]]),
          index but highest index
# str.join(iterable), Return a string
          which is the concatenation of
          the strings in iterable. A
          TypeError will be raised if
          there are any non-string values
           in iterable, including bytes
          objects. The separator between
          elements is the string
          providing this method.
", ".join(["spam", "spam", "spam"])  #
          'spam, spam, spam'
"-".join("Python")  # 'P-y-t-h-o-n'
"".join(["a", "b", "c"]) # 'abc'
"Hello World".lower()  # 'hello world',
          all lowercase
# str.lstrip(chars=None), Return a copy
          of the string with leading
          characters removed. The chars
          argument is a string specifying
          the set of characters to be
          removed. If omitted or None,
          the chars argument defaults to
          removing whitespace. The chars
          argument is not a prefix;
          rather, all combinations of its
          values are stripped.
"   spacious   ".lstrip()  # 'spacious
          '
"www.example.com".lstrip("cmowz.")  # '
          example.com'
# str.rstrip(chars=None), lstrip but
          trailing
# str.strip(chars=None), lstrip and
          rstrip
# str.partition(sep), Split the string
          at the first occurrence of sep,
           and return a 3-tuple
          containing the part before the
          separator, the separator itself
          , and the part after the
          separator. If the separator is
          not found, return a 3-tuple
          containing the string itself,
          followed by two empty strings.
"Monty Python".partition(" ")  # ('
          Monty', ' ', 'Python')
"Monty Python's Flying Circus"
          .partition(" ")  # ('Monty', '
          ', "Python's Flying Circus")
"Monty Python".partition("-")  # ('
          Monty Python', '', '')
# str.rpartition(sep), partition but
          last occurrence
# str.removeprefix(prefix), If the
          string starts with the prefix
          string, return string[len(
          prefix):]. Otherwise, return a
          copy of the original string:
"Arthur: three!".removeprefix("Arthur:
          ") # 'three!'
# str.removesuffix(suffix, /), If the
          string ends with the suffix
          string and that suffix is not
          empty, return string[:-len(
          suffix)]. Otherwise, return a
          copy of the original string
# str.replace(old, new, count=-1),
          Return a copy of the string
          with all occurrences of
          substring old replaced by new.
          If count is given, only the
          first count occurrences are
          replaced. If count is not
          specified or -1, then all
          occurrences are replaced.
"spam, spam, spam".replace("spam", "
          eggs")  # 'eggs, eggs, eggs'
"spam, spam, spam".replace("spam", "
          eggs", 1)  # 'eggs, spam, spam'
# str.split(sep=None, maxsplit=-1)
          Return a list of the words in
          the string, using sep as the
          delimiter string. If maxsplit
          is given, at most maxsplit
          splits are done, the rightmost
          ones (thus, the list will have
          at most maxsplit+1 elements).
          If maxsplit is not specified or
           -1, then there is no limit on
          the number of splits (all
          possible splits are made). If
          sep is given, consecutive
          delimiters are not grouped
          together and are deemed to
          delimit empty strings (for
          example, '1,,2'.split(',')
          returns ['1', '', '2']). The
          sep argument may consist of
          multiple characters as a single
           delimiter. Splitting an empty
          string with a specified
          separator returns ['']. If sep
          is not specified or is None, a
          different splitting algorithm
          is applied: runs of consecutive
           whitespace are regarded as a
          single separator, and the
          result will contain no empty
          strings at the start or end if
          the string has leading or
          trailing whitespace.
          Consequently, splitting an
          empty string or a string
          consisting of just whitespace
          with a None separator returns
          [].
"1,2,3".split(",")  # ['1', '2', '3']
"1,2,3".split(",", maxsplit=1)  # ['1',
          '2,3']
"1,2,,3,".split(",")  # ['1', '2', '',
          '3', '']
"1<>2<>3<4".split("<>")  # ['1', '2',
          '3<4']
# str.rsplit(sep=None, maxsplit=-1),
          split but rightmost
# str.swapcase()
"Hello World".swapcase()  # 'hELLO
          wORLD'
# str.title(), Return a titlecased
          version of the string where
          words start with an uppercase
          character and the remaining
          characters are lowercase. The
          algorithm uses a simple
          language-independent definition
           of a word as groups of
          consecutive letters.
"Hello world".title()  # 'Hello World'
"they're bill's friends from the UK".
          title()  # "They'Re Bill'S
          Friends From The Uk"
# str.upper(), all uppercase




# A list is a dynamic array of elements
          of arbitrary types
a = [10, 20, 30]
type(a)  # <class 'list'>
a[0]  # 10, O(1)
a[1]  # 20, O(1)
a[-1]  # 30, O(1)
a[-2]  # 20, O(1)
len(a)  # 3, O(1)
a.__len__()  # 3, O(1)
a.append(40)  # a == [10, 20, 30, 40],
          amortized O(1)
a.pop()  # a == [10, 20, 30], O(1)
a.insert(1, 50)  # a == [10, 50, 20,
          30], O(n)
a.pop(0)  # a == [50, 20, 30], O(n)
a[0] = 0  # a == [0, 20, 30], O(1)
b, c, d = a  # b == 0, c == 20, d == 30
a.clear()  # a == []
a = [0, 1, 2]
b = a  # id(a) == id(b)
b.pop()  # a == b == [0, 1], id(a) ==
          id(b)
b = [1, 2]  # id(a) != id(b), a == [0,
          1]
a = [1, 1, 1]
a.count(1)  # 3, O(n)
b = a.copy()  # a == b but id(a) != id(
          b)
a.index(1)  # 0, first occurrence, O(n)
a.extend([0, 4, 3, 4])  # a == [1, 1,
          1, 0, 4, 3, 4], O(k)
a.remove(4)  # a.pop(a.index(x))
a.reverse()  # a == [4, 3, 0, 1, 1, 1],
          O(n)
a.sort()  # [0, 1, 1, 1, 3, 4], O(n log
          n) worst-case O(n) best-case,
          Powersort
sorted(a)  # same as a.sort()
del a[0:5:2]  # a == [1, 1, 4]
1 in a  # True
0 not in a  # True
a = [None] * 5  # a == [None, None,
          None, None, None]
del a  # a is now not defined
a = [1, "1", [1], True]  # arbitrary
          types
a.__getitem__(0)  # 1, same as a[0]
a.__setitem__(0, (1,))  # a == [(1,),
          '1', [1], True], same as a[0] =
          (1,)
it = iter([1, 2, 3])  # also it = [1,
          2, 3].__iter__()
iter(a) is iter(a)  # False
next(it)  # 1, also it.__next__()
# Return of __next__ is defined by the
          yield clause in the method.
# __next__ is used by for loop.
# range(start, stop, step), all are
          integers, start + k step, k =
          [0, (stop - start) / step - 1]
list(range(0, 10, 1))  # [0, 1, 2, 3,
          4, 5, 6, 7, 8, 9]
list(range(9, -1, -2))  # [9, 7, 5, 3,
          1]
# List comprehensions provide a concise
          way to create lists. Common
          applications are to make new
          lists where each element is the
          result of some operations
          applied to each member of
          another sequence or iterable,
          or to create a subsequence of
          those elements that satisfy a
          certain condition.
list(map(lambda x: x**2, range(10)))  #
          [0, 1, 4, 9, 16, 25, 36, 49,
          64, 81]
[x**2 for x in range(10)]  # [0, 1, 4,
          9, 16, 25, 36, 49, 64, 81]
```

```
[(x, y) for x in [1, 2, 3] for y in [3,
          1, 4] if x != y]  # [(1, 3),
          (1, 4), (2, 3), (2, 1), (2, 4),
          (3, 1), (3, 4)]
[w.strip() for w in [" banana", "
          loganberry ", "passion fruit  "
          ]]  # ['banana', 'loganberry',
          'passion fruit']
[num for elem in [[1, 2, 3], [4, 5, 6],
          [7, 8, 9]] for num in elem]  #
          [1, 2, 3, 4, 5, 6, 7, 8, 9]
[
    [
        row[i]
        for row in [
            [1, 2, 3, 4],
            [5, 6, 7, 8],
            [9, 10, 11, 12],
        ]
    ]
    for i in range(4)
]  # [[1, 5, 9], [2, 6, 10], [3, 7,
          11], [4, 8, 12]]
# zip(*iterables, strict=False),
          Iterate over several iterables
          in parallel, producing tuples
          with an item from each one, a
          similar to transposing a matrix
          . By default, zip() stops when
          the shortest iterable is
          exhausted. With strict=True, it
           raises a ValueError if one
          iterable is exhausted before
          the others.
list(
    zip(
        *[
            [1, 2, 3, 4],
            [5, 6, 7, 8],
            [9, 10, 11, 12],
        ]
    )
)  # [(1, 5, 9), (2, 6, 10), (3, 7,
          11), (4, 8, 12)]
list(zip(range(3), ["fee", "fi", "fo",
          "fum"]))  # [(0, 'fee'), (1, '
          fi'), (2, 'fo')]
# list(zip(range(3), ['fee', 'fi', 'fo
          ', 'fum'], strict=True))  #
          ValueError: zip() argument 2 is
           longer than argument 1
[0, *[1, 2, 3], 4, 5, 6]  # [0, 1, 2,
          3, 4, 5, 6]



# A tuple is an immutable ordered
          collection of arbitrary number
          of elements of arbitrary types
a = (
    1,
    "a",
    "a",
    [0],
)
type(a)  # <class 'tuple'>
a = 1, "a", "a", [0]  # can also be
          created without the parentheses
a[0]  # 1
a[-1]  # [0]
# a[0] = 0  # TypeError: 'tuple' object
          does not support item
          assignment
a[3][0] = 1  # allowed
a = ("a",)  # a == ('a',), a tuple. One
          -item tuple requires comma
a = ("a",)  # a == ('a',), a tuple
a = "a"  # a == 'a', a string
a = (1, 1, 1)
a.count(1)  # 3
a.index(1)  # 0, first occurrence
1 in a  # True
0 not in a  # True



# A set is an immutable unordered
          collection of arbitrary number
          of elements of hashable types
          with duplicates ignored
a = {
    1,
    "a",
    "a",
}  # t == {1, 'a'}
"a".__hash__()  # -8760825294024040742,
          a hashable type has __hash__
          method that returns an int
# a = {[0]} # TypeError: cannot use '
          list' as a set element (
          unhashable type: 'list')
type(a)  # <class 'set'>
type({})  # <class 'dict'>
a = {True, 1}  # t == {True}
a = {1, True}  # t == {1}
a = {False, 0}  # t == {False}
a = {0, False}  # t == {0}
len(a)  # 1
a.clear()  # a == set()
a = {1, 0, 2}  # a == {0, 1, 2}
b = a.copy()  # id(a) != id(b)
a.add(3)  # a == {0, 1, 2, 3}
a.remove(3)  # a == {0, 1, 2}, remove
          if present, KeyError if not
# a.remove(3)  # KeyError: 3
a.discard(2)  # a == {0, 1}, remove if
          present
a.discard(2)  # a == {0, 1}
a.pop()  # 0, a == {1}, Remove and
          return an arbitrary element
          from the set. Raises KeyError
          if the set is empty.
a.pop()  # 1
```

```
# a.pop  # KeyError: 'pop from an empty
          set'
a = {0, 1, 2, 3}
# arg is any iterable, denoted as b
          below
a.isdisjoint({0})  # False, a\cap b ==
          \varnothing
a.isdisjoint([0])
a.isdisjoint(0,)
a.isdisjoint("123")  # True, a\subseteq b
a <= a  # True, a\subseteq b
a < a  # False, a\subsetneq b
a.issuperset(a)  # True, a\supseteq b
a >= a  # True, a\supseteq b
a > a  # False, a\supsetneq b
a.union({4})  # {0, 1, 2, 3, 4}, a\cup
          b
a | {4}  # {0, 1, 2, 3, 4}, a\cup b
a.update({1})  # a == {0, 1, 2, 3},
          assign a\cup b to a
a |= {4}  # a == {0, 1, 2, 3, 4},
          assign a\cup b to a
a.intersection({3})  # {3}, a\cap b
a & {3}  # {3}, a\cap b
a & {6}  # set(), a\cap b
a.intersection_update(a)  # a == {0, 1,
          2, 3, 4}, assign a\cap b to a
a &= {-1, 0, 1, 2, 3}  # a == {0, 1, 2,
          3}, assign a\cap b to a
a.difference({1})  # {0, 2, 3}, a\
          setminus b
a - {1}  # {0, 2, 3}, a\setminus b
a.difference_update({1})  # a == {0, 2,
          3}, assign a\setminus b to a
a -= {2}  # a == {0, 3}, assign a\
          setminus b to a
a.symmetric_difference({1, 3})  # {0,
          1}, a\cup b\setminus(a\cap b)
a ^ {1, 3}  # {0, 1}, a\cup b\setminus(
          a\cap b)
a.symmetric_difference_update({1, 3})
          # a == {0, 1}, assign a\cup b\
          setminus(a\cap b) to a
a ^= {1, 3}  # a == {0, 3}, assign a\
          cup b\setminus(a\cap b) to a
{x for c in "abracadabra" if c not in "
          abc"}  # {'d', 'r'}, set
          comprehension



# A dict is a mapping object maps
          hashable values to arbitrary
          objects
a = dict(one=1, two=2, three=3)
b = {"one": 1, "two": 2, "three": 3}
c = dict(zip(["one", "two", "three"],
          [1, 2, 3]))
d = dict([("two", 2), ("one", 1), ("
          three", 3)])
e = dict({"three": 3, "one": 1, "two":
          2})
f = dict({"one": 1, "three": 3}, two=2)
a == b == c == d == e == f  # True
type(d)  # <class 'dict'>
type({})  # <class 'dict'>
list(d)  # ['two', 'one', 'three'],
          list of keys
len(d)  # 3, len(list(d))
d["two"]  # 2
# d["a"]  # KeyError: 'a'
d["four"] = 5  # d == {'two': 2, 'one':
          1, 'three': 3, 'four': 5}
d["four"] = 4  # d == {'two': 2, 'one':
          1, 'three': 3, 'four': 4}
del d["four"]  # d == {'two': 2, 'one':
          1, 'three': 3}
# del d["five"]  # KeyError: 'five'
"one" in d  # True
"five" in d  # False
iter(d) is iter(d.keys())
e.clear()  # e == {}



class _SNode:
    def __init__(self, data, next=None
        ):
        self.data = data
        self.next = next



class SinglyLinkedList:
    def __init__(self):
        self.head = None
        self.tail = None
        self.size = 0

    def __len__(self):  # O(1)
        return self.size

    def is_empty(self):  # O(1)
        return self.head is None

    def __getitem__(self, num):  # O(
          num)
        if not 0 <= num < self.size:
            raise IndexError("list
                assignment index
                out of range")
        cur = self.head
        for _ in range(0, num):
            cur = cur.next
        return cur.data

    def __setitem__(self, num, x):  # O
          (num)
        if not 0 <= num < self.size:
            raise IndexError("list
                assignment index
                out of range")
        cur = self.head
        for _ in range(0, num):
            cur = cur.next
        cur.data = x

    def prepend(self, x):  # O(1)
        self.head = _SNode(x, self.head
            )
        if self.tail is None:
            self.tail = self.head
        self.size += 1

    def append(self, x):  # O(1)
        node = _SNode(x)
        if self.tail is None:
            self.head = self.tail =
                node
        else:
            self.tail.next = node
            self.tail = node
        self.size += 1

    def pop_front(self):  # O(1)
        if self.head is None:
            raise IndexError("remove
                from empty list")
        x = self.head.data
        self.head = self.head.next
        self.size -= 1
        if self.head is None:
            self.tail = None
        return x

    def peek_front(self):  # O(1)
        if self.is_empty():
            raise IndexError("peek at
                empty list")
        return self.head.data

    def peek_back(self):  # O(1)
        if self.is_empty():
            raise IndexError("peek at
                empty list")
        return self.tail.data

    def clear(self):  # O(1)
        self.head = None
        self.tail = None
        self.size = 0

    def __iter__(self):
        cur = self.head
        while cur is not None:
            yield cur.data
            cur = cur.next



class _DNode:
    def __init__(self, data, prev=None,
          next=None):
        self.data = data
        self.prev = prev
        self.next = next



class DoublyLinkedList:
    def __init__(self):
        self.head = None
        self.tail = None
        self.size = 0

    def __len__(self):  # O(1)
        return self.size

    def is_empty(self):  # O(1)
        return self.size == 0

    def __getitem__(self, num):  # O(
          min(n, size - n)), n = num
        if num > 0 else num + size
        if not -self.size <= num < self
            .size:
            raise IndexError("list
                assignment index
                out of range")
        if num * 2 >= self.size:
            num -= self.size
        elif num * 2 < -self.size:
            num += self.size
        if num >= 0:
            cur = self.head
            for _ in range(0, num):
                cur = cur.next
        else:
            cur = self.tail
            for _ in range(-1, num,
                -1):
                cur = cur.prev
        return cur.data

    def __setitem__(self, num, x):  # O
          (min(n, size - n)), n = num
        if num > 0 else num + size
        if not -self.size <= num < self
            .size:
            raise IndexError("list
                assignment index
                out of range")
        if num * 2 >= self.size:
            num -= self.size
        elif num * 2 < -self.size:
            num += self.size
        if num >= 0:
            cur = self.head
            for _ in range(0, num):
                cur = cur.next
        else:
            cur = self.tail
            for _ in range(-1, num,
                -1):
                cur = cur.prev
        cur.data = x

    def prepend(self, x):  # O(1)
        node = _DNode(x, None, self.
            head)
        if self.head is None:
            self.head = self.tail =
                node
        else:
            self.head.prev = node
            node.next = self.head
            self.head = node
        self.size += 1

    def append(self, x):  # O(1)
        node = _DNode(x, self.tail,
            None)
        if self.tail is None:
            self.head = self.tail =
                node
        else:
            self.tail.next = node
            self.tail = node
        self.size += 1

    def pop_front(self):  # O(1)
        if self.head is None:
            raise IndexError("pop from
                empty list")
        x = self.head.data
        self.head = self.head.next
        self.size -= 1
        if self.head is None:
            self.tail = None
        else:
            self.head.prev = None
        return x

    def pop_back(self):  # O(1)
        if self.tail is None:
            raise IndexError("pop from
                empty list")
        x = self.tail.data
        self.tail = self.tail.prev
        self.size -= 1
        if self.tail is None:
            self.head = None
        else:
            self.tail.next = None
        return x

    def pop_node(self, node):  # O(1)
        if self.head is None:
            raise IndexError("remove
                from empty list")
        if self.head.pop_front()
            is node:
            return self.pop_back()
        node.prev.next = node.next
        node.next.prev = node.prev
        self.size -= 1
        return node.data

    def peek_front(self):  # O(1)
        if self.is_empty():
            raise IndexError("peek at
                empty list")
        return self.head.data

    def peek_back(self):  # O(1)
        if self.is_empty():
            raise IndexError("peek at
                empty list")
        return self.tail.data

    def clear(self):  # O(1)
        self.head = None
        self.tail = None
        self.size = 0

    def __iter__(self):
        cur = self.head
        while cur is not None:
            yield cur.data
            cur = cur.next

class BlockListDeque:  # Both ends IO
          amortized O(1)
    def __init__(self, block_size=64,
          front_step=1, back_step=1):
        assert block_size > 0
        assert front_step > 0
        assert back_step > 0
        self.block_size = block_size
        self.front_step = front_step
        self.back_step = back_step
        self.data = [None] * self.
            block_size for _ in
            range(self.front_step +
            self.back_step - 1)]
        self.block_begin = 0
        self.block_end = self.
            front_step
        self.in_begin = (self.
            block_size - 1) // 2
        self.in_end = self.in_begin
        self.size = 0

    def __len__(self):
        return self.size

    def is_empty(self):
        return self.size == 0

    def capacity(self):
        return self.block_size * len(
            self.data)

    def is_valid(self):
        return self.size == self.
            block_size * (self.
            block_end - self.
            block_begin - 1) + self
            .in_end - self.in_begin

    def reserve_front(self, blocks=None
        ):
        if blocks is None:
            blocks = self.front_step
        new = [[None] * self.block_size
            for _ in range(blocks
            )]
        new.extend(self.data)
        self.data = new
        self.block_begin += blocks
        self.block_end += blocks

    def reserve_back(self, blocks=None
        ):
        if blocks is None:
            blocks = self.back_step
        self.data.extend([[None] * self
            .block_size for _ in
            range(blocks)])

    def shrink_front_to_fit(self):
        new = self.data[self.
            block_begin :]
        self.data = new
        self.block_end -= self.
            block_begin
        self.block_begin = 0

    def shrink_back_to_fit(self):
        for i in range(self.block_end,
            len(self.data)):
            self.data.pop()

    def prepend(self, x):
        if self.in_begin != 0:
            self.in_begin -= 1
        else:
            if self.block_begin == 0:
                self.reserve_front()
            self.block_begin -= 1
            self.in_begin = self.
                block_size - 1
            self.data[self.block_begin][
                self.in_begin] = x
        self.size += 1

    def append(self, x):
        if self.in_end != self.
            block_size:
            self.in_end += 1
        else:
            if self.block_end == len(
                self.data):
                self.reserve_back()
            self.block_end += 1
            self.in_end = 1
        self.data[self.block_end - 1][
            self.in_end - 1] = x
        self.size += 1

    def pop_front(self):
        if self.is_empty():
            raise IndexError("pop from
                empty deque")
        x = self.data[self.block_begin
            ][self.in_begin]
        self.data[self.block_begin][
            self.in_begin] = None
        self.size -= 1
        if self.in_begin != self.
            block_size - 1:
            self.in_begin += 1
        else:
            self.block_begin += 1
            self.in_begin = 0
        return x

    def pop_back(self):
        if self.is_empty():
            raise IndexError("pop from
                empty deque")
        x = self.data[self.block_end -
            1][self.in_end - 1]
        self.data[self.block_end - 1][
            self.in_end - 1] = None
        self.size -= 1
        if self.in_end != 1:
            self.in_end -= 1
        else:
            self.block_end -= 1
            self.in_end = self.
                block_size
        return x

    def peek_front(self):
        if self.is_empty():
            raise IndexError("peek from
                empty deque")
        return self.data[self.
            block_begin][self.
            in_begin]

    def peek_back(self):
        if self.is_empty():
            raise IndexError("peek from
                empty deque")
        return self.data[self.block_end
            - 1][self.in_end - 1]

    def clear(self):
        for _ in range(self.size):
            self.pop_back()

    def __iter__(self):
        block = self.block_begin
        index = self.in_begin
        for _ in range(self.size):
            yield self.data[block][
                index]
            index += 1
            if index == self.block_size
                :
                index = 0
                block += 1
```